



Fatemeh Aghaei

Nationality: Dutch, Iranian

MSc Bioinformatics / Laboratory Digitalization / GxP Automation & Data Analysis

Working Experiences

- 2022–Present **Senior Microbiology Technician**, *Johnson & Johnson Innovative Medicine, QC-Microbiology Department*, Leiden, the Netherlands.
- Support GMP microbiology and CAR-T manufacturing operations through laboratory testing, data review, trend reporting, investigation support, and compliant documentation.
 - Contribute to laboratory digitalization and continuous improvement projects by translating QC workflow needs into reporting tools, automated workflows, and data visualizations.
 - Initiated an automated QC reporting tool concept that was shared with global J&J stakeholders, using structured data, API/database connections, and automation concepts to improve reporting efficiency, data accessibility, and GMP compliance
- 2026–Present **Volunteer Data Analyst**, *Statistics Without Borders*.
- Clean, curate, and validate large structured healthcare datasets in R, supporting reproducible analysis, visualization, and communication for a child survival and hospitalization risk study.
- 2019 **Microbiology Laboratory Technician**, *Naft Hospital*, Tehran, Iran.
- Performed routine microbiological testing in a clinical laboratory, supporting accurate result documentation and reporting.

Education and training

- 2023–2025 **Master's degree, Bioinformatics**, *University of Birmingham*, Birmingham, United Kingdom.
- Grade: Distinction | Graduated: Dec 2025
- Applied R, Python, and machine learning concepts to biomedical data analysis, building reproducible workflows for data processing, integration, visualization, and reporting.
 - Thesis project: investigated ARNT as a prognostic biomarker in non-muscle-invasive bladder cancer, integrating molecular, clinical, survival, and pathway-level data.
- 2017 - 2021 **Bachelor's degree, Cellular and Molecular Biology - Microbiology**, *Azad University, Tehran Medical Branch*, Tehran, Iran.
- Built a strong foundation in microbiology, molecular biology, genetics, and biomedical laboratory science.

Technical Skills

Programming & Platforms: R, Python, Git/GitHub, Linux command line, Quarto, LaTeX, Galaxy, Databricks

Lab Systems: GMP/GxP documentation, QC data review, trend reporting, workflow automation, reporting tools, API/database connections, structured data handling

Data Analysis: data wrangling, statistical analysis, visualization, reproducible reporting, RNA-seq, CNV analysis, pathway enrichment, survival analysis, biomarker analysis

Selected Courses

Present **Algorithms and Data Structures**, Offered by Princeton University. Passing on Coursera.org.

2023-2024 **Health Data Analytics**, Offered by University of Birmingham, part of Bioinformatics Master's degree, Focus on healthcare data acquisition, integration, visualization, and statistical analysis.

2022 **Introduction to Probability and Data with R (Certificate)**, Offered by Duke University. Passed on Coursera.org.

2023-2024 **Statistical Machine Learning**, Offered by University of Birmingham, part of Bioinformatics Master's degree, Application of machine learning models for data-driven insights, predictive analytics, and decision-making support.

2023-2024 **Applied Genomics and Next Generation Sequencing**, Offered by University of Birmingham, part of Bioinformatics Master's degree, Focus on data processing, integration, and analysis of large-scale genomic data from sequencing technologies.

2024-2025 **Applied Metabolomics and Advanced (Omics) Technologies**, Offered by University of Birmingham, part of Bioinformatics Master's degree, Focus on metabolomics data processing, integration, and multi-omics applications.

Volunteering Experiences

2022-Present **Volunteer**, Stichting NL Cares, Amsterdam, the Netherlands.

Languages

English (Fluent), Dutch (Intermediate), Persian (Native)